Interaction between fluids and Helfrich membranes

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I. INTRODUCTION

A. State of the art

A number of existing studies rely on the hypothesis that the bulk fluid surrounding the Helfrich membrane is inertialess. For instance, Arroyo et al. study the interaction between a Helfrich membrane and bulk fluid in the Stokes approximation, for simple membrane geometries that can be described by a single parameter [?]. Boedec and coauthors combined finite element and boundary element methods describe an Helfrich membrane interacting with an inertialess bulk fluid [?]. In there, the membrane surface rheology, such as viscous effects and flows induced by surface-tension gradients, is neglected. A Helfrich membrane coupled to a bulk fluid has been recently studied by Zhou and co-authors with boundary element and finite element methods, under the assumption that the bulk fluid is inertialess and that Helfrich membrane is inextensible [?]. Barakat and Shaqfeh modelled vesicles in a circular tube by means of boundary element methods and lubrication theory, by neglecting Helfrich membrane rheology and describing bulk fluid as inertialess [?]. Finally, Rodrigues et al., studied the interaction between an inertialess, closed Helfrich membranes and bulk fluids with finite element methods, by adopting a minimal description for the fluid, which is solely modelled as a volume constraint [?].

Other studies go beyond the inertialess approximation, and rely on simplifying assumptions for the Helfrich membrane. For instance, Bonito et al. propose a finite element model for a Helfrich membrane interacting an bulk fluid, the latter being in the inertial regime [? ?]. In this study Helfrich membrane is treated as a purely elastic body, characterized solely by bending rigidity, area and volume constraints—its fluid features, such as tangential and normal flows, are neglected. Laadhari extended the analyses above to the interaction between a Helfrich membrane and a non-Newtonian bulk fluid, in which membrane rheology is still neglected [?]. The interaction between a Helfrich membrane and a bulk fluid—both in the inertial regime—has been studied by Barrett and co-authors, [?], under the assumption that the Helfrich membrane is perfectly inextensible. Such analysis is based on a fluidic approach, in which the Helfrich membrane does not obey its own, intrinsic rheology—see below—but it is advected by the surrounding bulk fluid. As a result, the membrane tangential and normal velocities are not determined as the result of the surfacial Navier-Stokes equations, but they match the respective components of the bulk fluid velocity.

In this study, we propose a novel finite element study of the interaction between a bulk fluid (F) and a Helfrich membrane (M), based on the arbitrary Lagrangian-Eulerian method [? ?]. Our analysis includes a complete description of the physical features of both Helfrich membrane and bulk fluid, it may describe the inertial regime of both, and it accounts for a detailed description of Helfrich membrane elasticity, rheology, and extensibility. Unlike fluidic approaches where the Helfrich membrane dynamics is determined by the bulk fluid, here both the flows and shape of Helfrich membrane are determined by intrinsic physical evolution equations. In addition, our study can account for both closed and open surfaces, by leveraging the flexibility of its finite element formulation [?].

We solve for the system rheology by implementing a splitting scheme [? ?] for the Navier-Stokes equations, applied simultaneously to the Helfrich membrane and bulk fluid flow dynamics, and demonstrate it to be numerically stable. As for the membrane shape, we generalize the arc length gauge—previously proposed to describe the steady state of membrane tubes [?]—to the system

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dynamics, by means of a time-dependent gauge fixing. Overall, this implementation allows for handling strongly turbulent Helfrich membrane and bulk fluid flows, as well as large Helfrich membrane deformations.

The paper is structured as follows. In Section II we discuss the arbitrary Lagrangian-Eulerian method, and present two illustrative examples: The interaction between bulk fluid and a rigid body in Section II A, and that between bulk fluid and elastic body in Section II B. In Section II C we present our approach for the interaction between bulk fluid and Helfrich membrane. In particular, in Section II C 1 and ????, we discuss the splitting scheme and gauge-fixing procedure for the Helfrich membrane, bulk fluid domain and bulk fluid, respectively. Finally, ?? is devoted to the discussion of the results, and future directions.

II. RESULTS

In all systems that we will consider, we will introduce the reference and current configuration [? ? ?], see Fig. 2. All quantities relative to the reference and current configuration will be denoted by the superscript reference and current, respectively. In the reference configuration, the region occupied by bulk fluid is described by Cartesian coordinates $\mathbf{x}_{\mathrm{r}}$. In the current configuration, the region occupied by bulk fluid is described by Cartesian coordinates $\mathbf{x}_{\mathrm{c}}$.

The reference configuration is related to the current one as follows [?], see Fig. 2. At time t, the point $\mathbf{x}_{\mathrm{c}}$ corresponds to a point $\mathbf{x}_{\mathrm{r}}$ in the reference configuration through the deformation field [?]

$$\boldsymbol{u}_{\mathrm{D}}^{t}(\mathbf{x}_{\mathrm{r}}) \equiv \mathbf{x}_{\mathrm{c}} - \mathbf{x}_{\mathrm{r}}. \tag{1}$$

and the deformation-gradient tensor

$$F_{\alpha\beta}(\boldsymbol{u}) \equiv \delta_{\alpha\beta} + \frac{\partial u^{\alpha}}{\partial \mathrm{x}_{\mathrm{r}}^{\beta}}. \tag{2}$$

In what follows, greek indices $\alpha, \beta, \cdots$ will be used to denote vectors and tensors in two-dimensional Euclidean space—their position as upper or lower indices is thus immaterial [? ?].

We will characterize the mapping between $\mathbf{x}_{\mathrm{c}}$ and $\mathbf{x}_{\mathrm{r}}$ configurations in Eq. (1) with the fields

$$\mathbf{x}_{\mathrm{c}} = \boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), \tag{3}$$
$$\mathbf{x}_{\mathrm{r}} = \boldsymbol{\psi}^{t}(\mathbf{x}_{\mathrm{c}}). \tag{4}$$

We denote the bulk fluid velocity and surface tension by $v_{\mathrm{F\,c}}(\mathbf{x}_{\mathrm{c}})$ and $\varsigma_{\mathrm{F}}^{\mathrm{c}}(\mathbf{x}_{\mathrm{c}})$, respectively, where the superscript current specifies that these fields depend on the coordinate $\mathbf{x}_{\mathrm{c}}$ in the current configuration.

For each field $\boldsymbol{v}_{\mathrm{F\,c}}(\mathbf{x}_{\mathrm{c}}), \varsigma_{\mathrm{F}}(\mathbf{x}_{\mathrm{c}})$ which depends on the coordinate $\mathbf{x}_{\mathrm{c}}$ in the current configuration, we introduce its counterpart in the reference configuration:

$$\boldsymbol{v}_{\mathrm{F\,r}}(\mathbf{x}_{\mathrm{r}}, t) \equiv \boldsymbol{v}_{\mathrm{F\,c}}(\boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), t), \tag{5}$$
$$\varsigma_{\mathrm{F}}^{\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t) \equiv \varsigma_{\mathrm{F}}^{\mathrm{c}}(\boldsymbol{\varphi}^{t}(\mathbf{x}_{\mathrm{r}}), t). \tag{6}$$

The solution of the interaction dynamics between bulk fluid and rigid body, elastic body or Helfrich membrane is involved because of the presence of a moving boundary, cf. Figs. 2 and 3 and ??, respectively. Following the arbitrary Lagrangian-Eulerian procedure, we will bridge between the Lagrangian and description used to describe rigid body, elastic body or Helfrich membrane, and the Eulerian description of bulk fluid [?]. To achieve this, in what follows we will consider the equations of motion for rigid body, elastic body or Helfrich membrane and bulk fluid—which are naturally formulated the current configuration—and rewrite them in the reference configuration.

A. Bulk fluid and rigid body

In this Section, we will discuss the interaction between a bulk fluid and the simplest structure—a rigid body which is only allowed to turn about one point. Only the main results will be presented here; details can be found in ??????, respectively, and the temporal discretization is discussed in ??.

As shown in Fig. 2, in the reference configuration the axis of the ellipse, e..g, rigid body, lies parallel to the $\mathrm{x}_{\mathrm{c}}^{1}$ axis, while in the current configuration, such axis forms an angle θ (red arc) with respect to the $\mathrm{x}_{\mathrm{c}}^{1}$ axis,

An example of the system dynamics is shown in Fig. 1, where we study the interaction between air (bulk fluid) and an aerogel (rigid body)—an ultra light solid material involving a large porous structure [?]. The channel dimension are $L = 2.2\,\mathrm{m}$, $h = 0.41\,\mathrm{m}$, which have been taken from the Finite Element Analysis Tools 2D DFG 2D-3 benchmark example for a fuid flow around a cylinder [? ?], see Fig. 2. Given that the materials, such as bulk fluid and rigid body, are two dimensional, in what follows we will estimate their parameter values by considering their three-dimensional counterparts, and assume a unit thickness in the out-of-plane direction. The parameters for rigid body have been estimated from the properties of air: $\rho_{\mathrm{F}} = 1\,\mathrm{Kg/m^2}$, $\eta_{\mathrm{F}} = 10^{-5}\,\mathrm{Pa\,m\,sec}$ [?]. The inflow velocity profile is chosen as a Poiseuille flow

$$\boldsymbol{v}_{\mathrm{F}\,\square}(\mathbf{x}_{\mathrm{c}}) = \frac{6 v_0}{h^2} \mathrm{x}_{\mathrm{c}}^2 (\mathrm{x}_{\mathrm{c}}^2 - h), \tag{7}$$

with $v_0 = 1\,\mathrm{m/\,sec}$. The bulk fluid domain mesh stiffness exponent has been chosen to be $\zeta = 3$ [?], see ?? for details. Finally, rigid body is an ellipse with semi-axes $a = 0.1\,\mathrm{m}$, $b = 0.075\,\mathrm{m}$, center located at $\mathbf{x}_{\mathrm{r}} = (0.25\,\mathrm{m}, 0.2\,\mathrm{m})$, and moment of inertia $I = 10^{-3}\,\mathrm{Kg\,m^2}$, which corresponds to an aerogel density of $\sim 6\,\mathrm{Kg/m^3}$ [?]. As shown in ??, rigid body has an oscillatory behavior, whose period can be understood with scaling arguments—see ??.

B. Fluid and elastic body

In this Section, we will build on the analysis of Section II A and put bulk fluid into interaction with a more complex physical object—an elastic body; only the main results will be presented here.

The reference and current configurations are shown in Fig. 3. Here, the elastic body boundary $\partial\Omega_{\bigcirc}$ may deform, but the inner elastic body boundary $\partial\Omega_{\odot}$ is pinned to the $\mathrm{x}_{\mathrm{c}}^1$, $\mathrm{x}_{\mathrm{c}}^2$ plane. The equations of motion are reported in ??, and the system dinamics is discussed in ??.

An example of the coupled dynamics of bulk fluid and elastic body is shown in Fig. 4, where we show the interaction of a bulk fluid with a hydrogel (elastic body). Here, $\partial\Omega^{\mathrm{r}}_{\bigcirc}$ is an ellipse with semi-axes $a = 0.2\,\mathrm{m}$, $b = 0.1\,\mathrm{m}$ and center located at $\mathbf{x}_{\mathrm{r}} = (0.4\,\mathrm{m}, 0.2\,\mathrm{m})$, and $\partial\Omega^{\mathrm{r}}_{\odot}$ has radius $r = 0.0125\,\mathrm{m}$ and center located at $\mathbf{x}_{\mathrm{r}}$. The channel dimensions and bulk fluid parameters have been taken from the Finite Element Analysis Tools 2D DFG 2D-3 benchmark example for a fluid flow around a cylinder [? ?]. The elastic body density is $\rho_{\mathrm{E}} = 10^3\,\mathrm{Kg/m^2}$ [?]. The storage modulus, which approximates the shear modulus μ [?] is $\sim 10\,\mathrm{Kg/\,sec^2}$ [?]. Given a elastic body Poisson ratio [?] of $\nu \sim 0.49$ [?], we obtain $K \sim 5 \times 10^2\,\mathrm{Kg/\,sec^2}$ from the relation $K = 2\mu(1+\nu)/[3(2\nu - 1)]$ [?]. The bulk fluid inflow velocity profile is a Poiseuille flow, Eq. (7), with $v_0 = 1\,\mathrm{m/\,sec}$. Finally, the bulk fluid domain mesh stiffness exponent has been chosen to be $\zeta = 3$ [?].

C. Bulk fluid and Helfrich membrane

In this Section, we will build on the analysis of Section II B, and complexify the structure of the elastic body with which bulk fluid interacts by considering, instead of an elastic material, an Helfrich membrane [? ?].

For this problem, we will consider the geometry depicted in ??. A vertical channel with mirror symmetry about its mid axis is considered. The left boundary of the channel is $\partial\Omega_{\square}$, and the right boundary $\partial\Omega_{\square}$ lies on the symmetry axis. Fluid is injected from bottom to top on $\partial\Omega_{\square}$. Here, Helfrich membrane lies on the top boundary of bulk fluid, $\partial\Omega_{\square}$, which coincides with the Helfrich membrane manifold $\partial\Omega_{\_}$. Finally, we denote by Ω the region enclosed by the boundaries, and all the quantities above will be considered in the reference and current configurations.

The geometry of ?? may represent bulk fluid being confined in a large reservoir, lying on bottom of the channel, where the channel is a vertical neck of the reservoir, covered on top by Helfrich membrane.

1. *Helfrich membrane motion*

Both the physical nature and mathematical description of Helfrich membrane is significantly more complex than that of elastic body. First, unlike elastic body, Helfrich membrane has a fluid structure: its material elements can flow both tangentially and normally to it. Such fluid behavior requires an Eulerian description, which then needs to be connected to the Lagrangian description used to describe bulk fluid domain. Second, Helfrich membrane is described by fourth-order partial differential equations [?], which are significantly more complex than the second-order partial differential equations which describe elastic body [?]. Third, the geometrical description of Helfrich membrane requires fixing an appropriate *gauge*. In fact, a general parametrization of Helfrich membrane is given by two functions $\mathbf{x}_{\mathrm{c}}^{\mathrm{M}}(x^1)$. Both of these functions depend on the curvilinear coordinate x^1 which is defined in the reference domain of Helfrich membrane, $\partial\Omega^{\mathrm{r}}_{\_}$, see ??—note that here x^i differs from the Cartesian coordinate x^α. Despite this pair of functions, a single partial differential equation determines the Helfrich membrane shape [?]. To understand this redundancy, consider the stretching field $\nu > 0$, defined by

$$\nu^2 \equiv \boldsymbol{e}_1 \cdot \boldsymbol{e}_1, \tag{8}$$

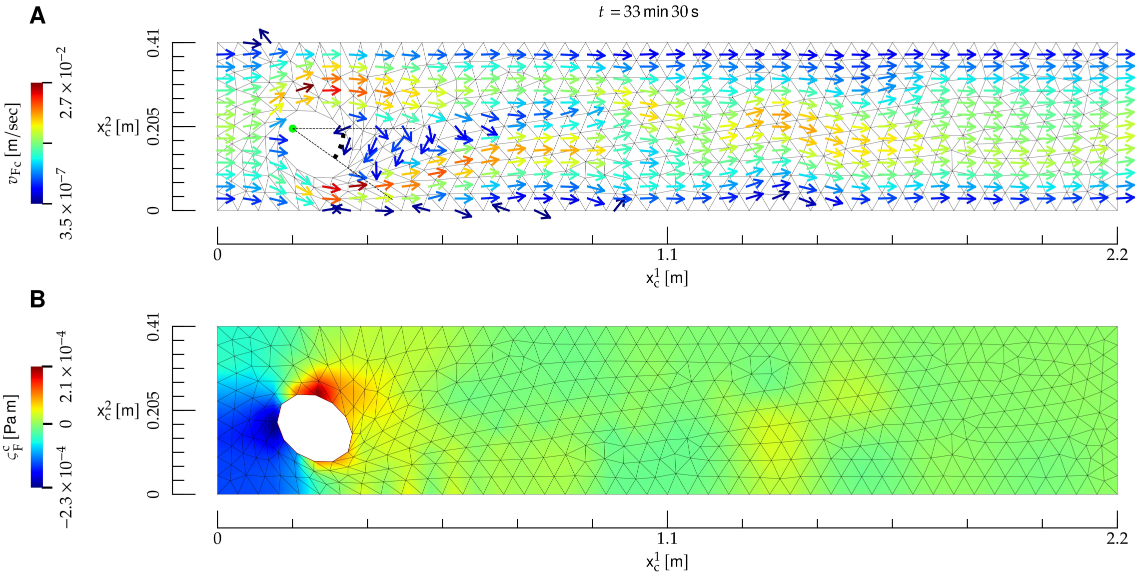

FIG. 1: Interaction between a bulk fluid (F) and a rigid rigid body (R), whose material properties roughly correspond to air (bulk fluid) and an aerogel (rigid body). **A**) Temporal snapshot of the mesh (black lines) and velocity field (arrows) at the current time t, shown on top. The direction of the velocity field is indicated by the arrows, and its norm by their color; the direction of arrows with velocity close to zero is not defined. Here, rigid body (ellipse) is allowed to pivot about its left focal point (red dot), and the related angle with respect to the $\mathrm{x}_{\mathrm{c}}^{1}$ direction is denoted by a red arc. Here, the Reynolds number is $R \sim 2 \times 10^{2}$. **B**) Color map of the surface tension at the current instant of time t.

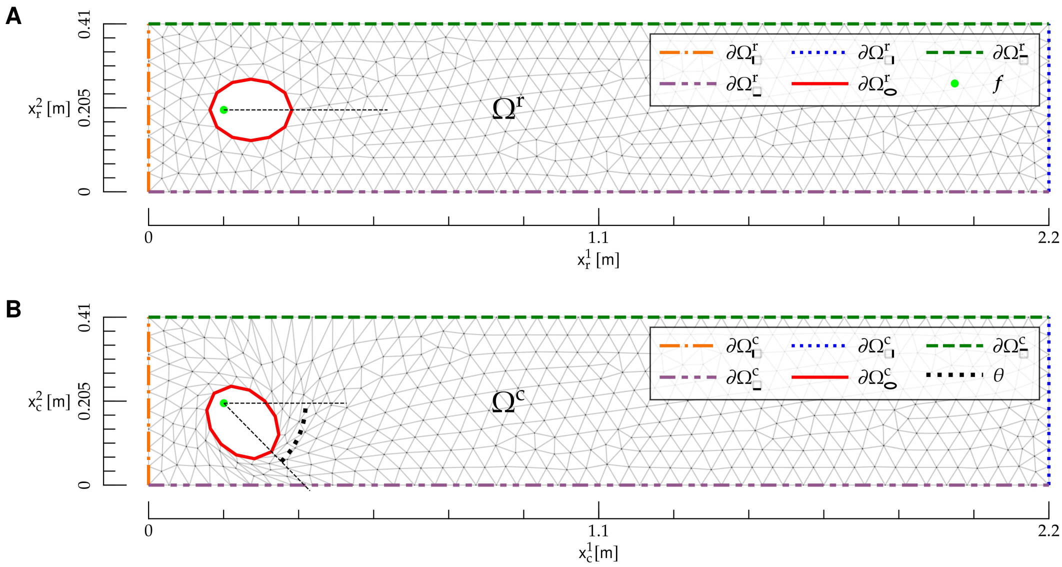

FIG. 2: Reference (r) and current (c) configuration for the interaction between a bulk fluid and a rigid body (R). **A**) reference configuration. The region Ω^{r} is given by the rectangle $0 \le \mathrm{x}_{\mathrm{r}}^{1} \le L$, $0 \le \mathrm{x}_{\mathrm{r}}^{2} \le h$ with the elliptical hole (rigid body), and is delimited by the mesh (gray lines). Boundaries in the reference configuration are denoted by $\partial\Omega^{\mathrm{r}}_{\square}$, $\partial\Omega^{\mathrm{r}}_{\square}$, $\partial\Omega^{\mathrm{r}}_{\square}$, $\partial\Omega^{\mathrm{r}}_{\square}$ and $\partial\Omega^{\mathrm{r}}_{\circ}$ (colored dashed lines). The ellipse focal point $\boldsymbol{f}$ (light green dot) is also shown. **B**) current configuration. The region Ω^{c} is delimited by the mesh (gray lines). Boundaries in the current configuration are denoted by $\partial\Omega^{\mathrm{c}}_{\square}$, $\partial\Omega^{\mathrm{c}}_{\square}$, $\partial\Omega^{\mathrm{c}}_{\square}$, $\partial\Omega^{\mathrm{c}}_{\square}$ and $\partial\Omega^{\mathrm{c}}_{\circ}$ (colored dashed lines), and the rotational angle θ of rigid body is shown as an arc (black dotted line). The Cartesian coordinates $\mathbf{x}_{\mathrm{c}}$ are also shown.

$$\boldsymbol{e}_i \equiv \frac{\partial \mathbf{x}_{\mathrm{c}}^{\mathrm{M}}}{\partial x^i}. \tag{9}$$

It is then easy to realize that two parametrizations with different νs may yield the same physical shape.

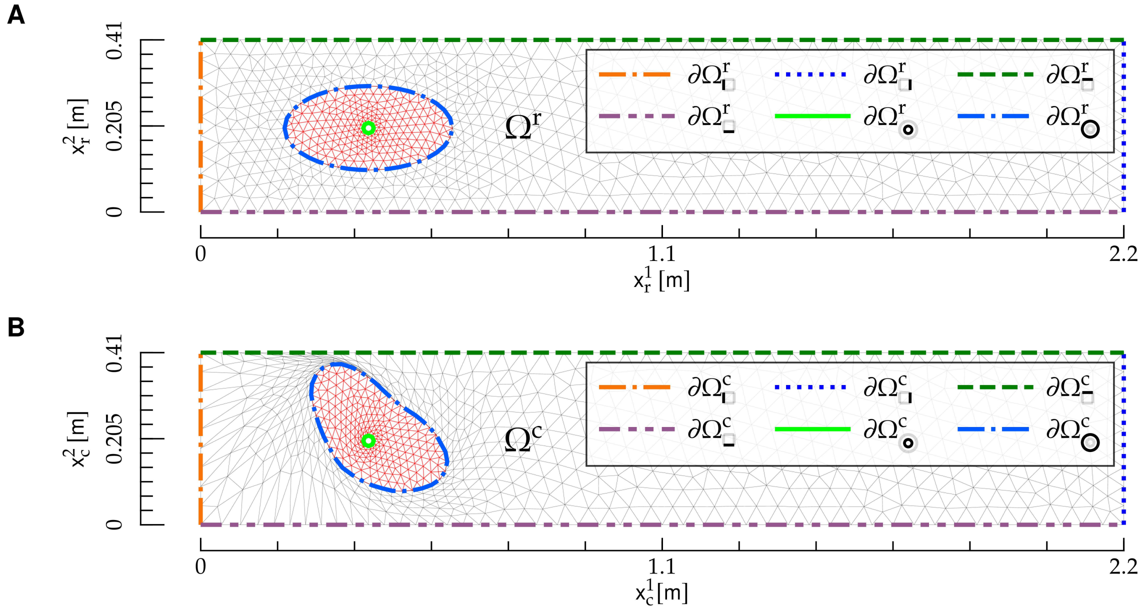

FIG. 3: Reference and current configuration for the interaction between a bulk fluid and an elastic body. The figures follows the same notation as Fig. 2.

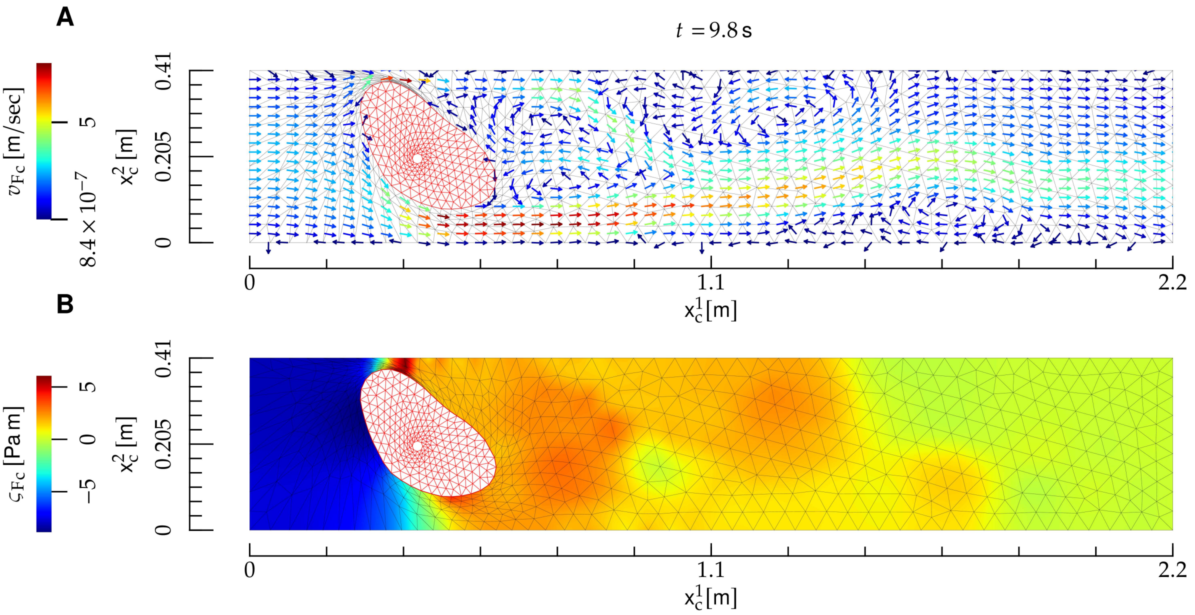

FIG. 4: Interaction between a bulk fluid and an elastic body (E) (a hydrogel). The notation is the same as in Fig. 1, and the elastic body is depicted as a red mesh. Here, the Reynolds number is $R \sim 2 \times 10^2$. The elastic body deformation is not symmetric under the mirror symmetry $\mathrm{x}_{\mathrm{c}}^2 \to h - \mathrm{x}_{\mathrm{c}}^2$, where h is the height of the bulk fluid domain, because the center of elastic body is not located at $\mathrm{x}_{\mathrm{c}}^2 = h/2$.

Such redundancy may be removed by a gauge fixing, i.e., by adding a constraint on $\mathbf{x}_{\mathrm{c}}^{\mathrm{M}}$, or its derivatives. Two common gauge choices are the Monge [**? ?**], or the arc-length parametrization [**?**], in which one sets

$$\nu = 1. \tag{10}$$

The arc-length gauge has been used to describe, for instance, the steady state for membrane tubules [**?**]. However, while a static constraint such as (10) is suited to static shapes, it cannot describe the Helfrich membrane dynamics, such as the one of Helfrich membrane interacting with bulk fluid. In fact, as Helfrich membrane is deformed, it will also stretch: as a result, Eq. (10) must be replaced by a time-dependent gauge—see below.

We describe Helfrich membrane with the Lagrangian approach, along the lines of the description of bulk fluid domain in Section II A and elastic body in Section II B. The Helfrich membrane displacement field is

$$\boldsymbol{u}_{\rm M}(x^1) \equiv \mathbf{x}_{\rm c}^{\rm M}(x^1) - \mathbf{x}_{\rm r}^{\rm M}(x^1), \tag{11}$$

where we choose the reference configuration as a straight line

$$\mathbf{x}_{\rm r}^{\rm M}(x^1) \equiv (x^1, h),\ 0 \leq x^1 \leq L, \tag{12}$$

which is shown in **??**A. In addition, we introduce the angle ψ formed by the tangent to Helfrich membrane with the x^1 axis [**?**], defined by

$$e_1^1 = \nu \cos\psi, \tag{13}$$
$$e_1^2 = -\nu \sin\psi. \tag{14}$$

Finally, the Helfrich membrane metric tensor is defined by [**?**]

$$g_{ij} \equiv \boldsymbol{e}_i \cdot \boldsymbol{e}_j. \tag{15}$$

In what follows, latin indices $i, j, k, \cdots$ will be used to denote vectors and tensors related to the Helfrich membrane manifold; their position as upper and lower indexes is thus important [**?**]. Because the differential manifold of Helfrich membrane is one dimensional, indexes $i, j, k, \cdots$ may take only one value, i.e., $i = j = k = \cdots = 1$. However, in formal relations such as Eq. (15), we will use the general notation $i, j, k, \cdots$ for the sake of generality, and for potential extensions of such relations to two-dimensional manifolds [**?**].

In order to work out the dynamical equations for Helfrich membrane, let us derive the force exerted by bulk fluid on Helfrich membrane. The force exerted on a line element $\mathrm{d}l_{\rm c}$ of Helfrich membrane is [**?**]

$$f_{{\rm F}\,\alpha}(\sigma_{\rm F}^{\rm c}, \nu, \psi) = -\sigma^{\rm c}_{{\rm F}\,\alpha\beta} \hat{n}^{{\rm c}\,\gamma}(\nu, \psi), \tag{16}$$

where

$$\sigma^{\rm c}_{{\rm F}\,\alpha\beta} \equiv \varsigma^{\rm c}_{\rm F}\delta_{\alpha\beta} + \eta_{\rm F}\left(\frac{\partial {v_{\rm F}}_{\rm c}^{\alpha}}{\partial \mathrm{x}_{\rm c}^{\beta}} + \frac{\partial {v_{\rm F}}_{\rm c}^{\beta}}{\partial \mathrm{x}_{\rm c}^{\alpha}}\right) \tag{17}$$

is the bulk fluid stress tensor [**?**] and $\hat{n}^{\rm c}$ the unit normal to $\partial\Omega^{\rm c}_{□}$ pointing outside $\Omega^{\rm c}$, see **??**. In Eq. (16), $\sigma^{\rm c}_{\rm F}$ is evaluated at the current coordinate $\mathbf{x}_{\rm c}^{\rm M} \in \partial\Omega^{\rm c}_{□}$ corresponding to x^1. The components of the force (16) tangential and normal to Helfrich membrane are [**?**]

$$f^i_{{\rm F}\,{\rm t}}(\sigma_{\rm F}^{\rm c}, \nu, \psi) = g^{ij}(\nu) e_j^{\alpha}(\nu, \psi) f_{\alpha}(\sigma_{\rm F}^{\rm c}, \nu, \psi) \tag{18}$$
$$f_{{\rm F}\,{\rm n}}(\sigma_{\rm F}^{\rm c}, \nu, \psi) = \hat{n}^{{\rm c}\,\alpha}(\nu, \psi) f_{\alpha}(\sigma_{\rm F}^{\rm c}, \nu, \psi), \tag{19}$$

where in Eq. (18) we used the feature that the metric tensor depends on ν only, see Eqs. (8) and (15).

The equations of motion for Helfrich membrane are [**? ? ?**]

$$\nabla_i v^i_{\rm M} - 2 H w_{\rm M} = 0, \tag{20}$$
$$\rho_{\rm M}\left(\frac{\partial v^i_{\rm M}}{\partial t} + v^j_{\rm M}\nabla_j v^i_{\rm M} - 2 v^j_{\rm M} w_{\rm M}\, b^i_j - w_{\rm M}\nabla^i w_{\rm M}\right) = \nabla^i \varsigma_{\rm M} + f^i_{\eta_{\rm M}\,{\rm t}} + f^i_{{\rm F}\,{\rm t}}, \tag{21}$$
$$\rho_{\rm M}\left[\frac{\partial w_{\rm M}}{\partial t} + v^i_{\rm M}(v^j_{\rm M} b_{ji} + \nabla_i w_{\rm M})\right] = 2\varsigma_{\rm M} H + f_{\kappa} + f_{\eta_{\rm M}\,{\rm n}} + f_{{\rm F}\,{\rm n}}, \tag{22}$$
$$\frac{\partial \boldsymbol{u}_{\rm M}}{\partial t} = w_{\rm M}\hat{n}^{\rm c}, \tag{23}$$
$$\frac{\partial u^1_{\rm M}}{\partial x^1} = -1 + \nu\cos\psi, \tag{24}$$
$$\frac{\partial u^2_{\rm M}}{\partial x^1} = -\nu\sin\psi, \tag{25}$$

and they hold for $x^1 \in \Omega^{\rm r}_{\_}$. Here, $v^i_{\rm M}$ and w are the Helfrich membrane tangential and normal velocity, respectively, ∇ is the covariant derivative related to the Levi-Civita connection associated to g, ∇_{LB} the Laplace-Beltrami operator, b the second fundamental form, and H and K the mean and Gaussian curvature, respectively. We refer to [? ? ? ?] for details on the differential-geometry formulation, and we will now discuss each line in Eqs. (20) to (25). Equation (20) enforces mass conservation [?]. Equations (21) and (22) are the tangential and normal components of the Navier-Stokes equations, respectively. The first term in the right-hand side of Eq. (21) represnts the force on membrane elements induced by gradients of line tension, while the first term in the right-hand side of Eq. (22) constitutes Laplace pressure [? ?]. The other terms read

$$
\begin{aligned}
f_\kappa(\nu, \psi) \equiv& \qquad (26)\\
-2\kappa[\nabla_i \nabla^i H + 2H(H^2 - K)],&\\
f^i_{\eta_{\mathrm{M}}\,\mathrm{t}}(v_{\mathrm{M}}, w_{\mathrm{M}}, \nu, \psi) \equiv& \qquad (27)\\
\eta_{\mathrm{M}}[-\nabla_{\mathrm{LB}} v^i_{\mathrm{M}} -&\\
2\left(b^{ij} - 2\,H\,g^{ij}\right)\nabla_j w_{\mathrm{M}} + 2K v^i_{\mathrm{M}}],&\\
f_{\eta_{\mathrm{M}}\,\mathrm{n}}(v_{\mathrm{M}}, w_{\mathrm{M}}, \nu, \psi) \equiv& \qquad (28)\\
2\eta_{\mathrm{M}}[(\nabla^i v^j_{\mathrm{M}}) b_{ij} - 2w_{\mathrm{M}}(2H^2 - K)].&
\end{aligned}
$$

In ??, f_κ is represents the resistance of Helfrich membrane to bending, and it is derived from Helfrich model [?], while $f_{\eta_{\mathrm{M}}\,t}$ and $f_{\eta_{\mathrm{M}}\,\mathrm{n}}$ in ???? are, respectively, the tangential and normal component of the viscous force resulting from in-membrane flows [?]. Equation (23) is the time-evolution equation for the Helfrich membrane shape, which evolves according to the normal $\hat{n}^c$, and it has been obtained from Eqs. (11) and (12). Equations (24) and (25) re-express the definitions of ν and ψ, and they result from Eqs. (9) and (11) to (14). Finally, ρ_{M}, ς_{M} and κ are, respectively, the density, line tension and bending rigidity of Helfrich membrane.

We choose the following boundary conditions for Eqs. (20) to (25):

$$v^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \qquad (29)$$

$$v^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \qquad (30)$$

$$w_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \qquad (31)$$

$$\nabla_i w_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \qquad (32)$$

$$\varsigma_{\mathrm{M}} = \varsigma_{\mathrm{M}\,\bullet\!-} \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \qquad (33)$$

$$\boldsymbol{u}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \qquad (34)$$

$$u^1_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \qquad (35)$$

$$\frac{\partial u^2_{\mathrm{M}}}{\partial x^1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}. \qquad (36)$$

?????? enforce the condition that the left extremity of Helfrich membrane, $\partial\Omega_{\bullet\!-}$, is clamped on the channel wall $\partial\Omega_{\square}$. ???????? enforce the mirror symmetry about $\partial\Omega_{\square}$. Namely, ?? ensures that the v_{M} vector field vanishes on $\partial\Omega_{-\!\bullet}$, because a nonzero value of this field on $\partial\Omega_{\square}$ would break the axial symmetry. The condition ?? follows the same lines. In addition, the symmetry requires that scalar fields and vertical components of vector fields in Ω have zero slope with respect to x^1 at $\partial\Omega_{-\!\bullet}$, see ????. Finally, ?? fixes the Helfrich membrane tension on $\partial\Omega_{\bullet\!-}$.

2. Bulk fluid domain motion

We will determine $\boldsymbol{u}_{\mathrm{D}}$ as the solution of a boundary-value problem given by the following elastic model [? ?]

$$\frac{\partial S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{D}})}{\partial \mathsf{x}^\beta_{\mathrm{r}}} = 0 \text{ in } \Omega^{\mathrm{r}}, \qquad (37)$$

where the elastic stress tensor S_{D} is given by

$$S_{\mathrm{D}\,\alpha\beta} \equiv F_{\alpha\gamma} T_{\gamma\beta}, \qquad (38)$$

and

$$
\begin{aligned}
T_{\alpha\beta} \equiv& K(\boldsymbol{u}_{\mathrm{D}}) E_{\gamma\gamma}\delta_{\alpha\beta} + \qquad (39)\\
& 2\mu(\boldsymbol{u}_{\mathrm{D}})\left(E_{\alpha\beta} - \frac{1}{2}\delta_{\alpha\beta}E_{\gamma\gamma}\right),\\
E_{\alpha\beta} \equiv& \frac{1}{2}\left(F_{\gamma\alpha}F_{\gamma\beta} - \delta_{\alpha\beta}\right), \qquad (40)\\
K(\boldsymbol{u}) \equiv& \frac{1}{[\det(F(\boldsymbol{u}))]^\zeta}, \qquad (41)\\
\mu(\boldsymbol{u}) \equiv& \frac{1}{[\det(F(\boldsymbol{u}))]^\zeta}. \qquad (42)
\end{aligned}
$$

In ????, we included an additional dependence of bulk modulus K and the shear modulus μ on the deformation-gradient tensor (2). In the absence of this dependence, the elastic model above would be unstable under local compression [?]. In fact, the model's Lagrangian would not penalize configurations with small $\det F$. The inclusion of the dependence ???? on $\det F$ allows for penalizing these configurations—the larger the exponent $\zeta > 0$, the stronger the penalty. As a result, this dependence makes the model stable under mesh compression, such as the one shown below rigid body in Fig. 2B.

The equation of motion for $\dot{\boldsymbol{u}}_{\mathrm{D}}$ is obtained by deriving ?? with respect to time, and using the fact that reference coordinates are time independent:

$$\frac{\partial}{\partial \mathsf{x}^\beta_{\mathrm{r}}} \frac{\mathrm{d}S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{D}})}{\mathrm{d}t} = 0 \text{ in } \Omega^{\mathrm{r}}. \qquad (43)$$

We impose the following boundary conditions for ????:

$$\boldsymbol{u}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square} \cup \partial\Omega^{\mathrm{r}}_{\square}, \tag{44}$$
$$u^1_{\mathrm{D}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{45}$$
$$\frac{\partial u^2_{\mathrm{D}}}{\partial \mathrm{x}^1_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{46}$$
$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{u}_{\mathrm{M}} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{47}$$
$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square} \cup \partial\Omega^{\mathrm{r}}_{\square}, \tag{48}$$
$$\dot{u}^1_{\mathrm{D}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{49}$$
$$\frac{\partial \dot{u}^2_{\mathrm{D}}}{\partial \mathrm{x}^1_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{50}$$
$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \dot{\boldsymbol{u}}_{\mathrm{M}} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}. \tag{51}$$

?? ensures that bulk fluid domain is not deformed on $\partial\Omega^{\mathrm{r}}_{\square} \cup \partial\Omega^{\mathrm{r}}_{\square}$, cf. **??**. **????** follow from axial symmetry, and **??** from the requirement that bulk fluid domain and Helfrich membrane deformations stay conforming [**?**]. Finally, the boundary conditions for $\dot{\boldsymbol{u}}_{\mathrm{M}}$, **????????**, are derived along the same lines as **????????**.

We observe that **??????????** imply that **????** contains up to second derivatives of $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$. As a result, the boundary conditions **????????????????** uniquely determine the solution of the boundary-value problem **????** [**?**].

3. *Bulk fluid motion*

We will now discuss the bulk fluid equations of motion, and present them first in current configuration coordinates, and then transform them in reference coordinates.

a. Equations of motion in current configuration coordinates In current coordinates, the bulk fluid equations of motion are the Navier-Stokes and continuity equations [**?**]:

$$\rho_{\mathrm{F}} \left(\partial_t v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}} + v_{\mathrm{F}}{}^{\beta}_{\mathrm{c}} \frac{\partial v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}}{\partial \mathrm{x}^{\beta}_{\mathrm{c}}} \right) = \frac{\partial \sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta}}{\partial \mathrm{x}^{\beta}_{\mathrm{c}}}, \tag{52}$$
$$\frac{\partial v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}}{\partial \mathrm{x}^{\alpha}_{\mathrm{c}}} = 0, \tag{53}$$

and they hold for $\mathbf{x}_{\mathrm{c}} \in \Omega^{\mathrm{c}}$. **????** describe the bulk fluid motion in a flat, two-dimensional geometry [**?**]—an example of the corresponding Navier-Stokes equations on a curved geometry is given by Eq. (21).

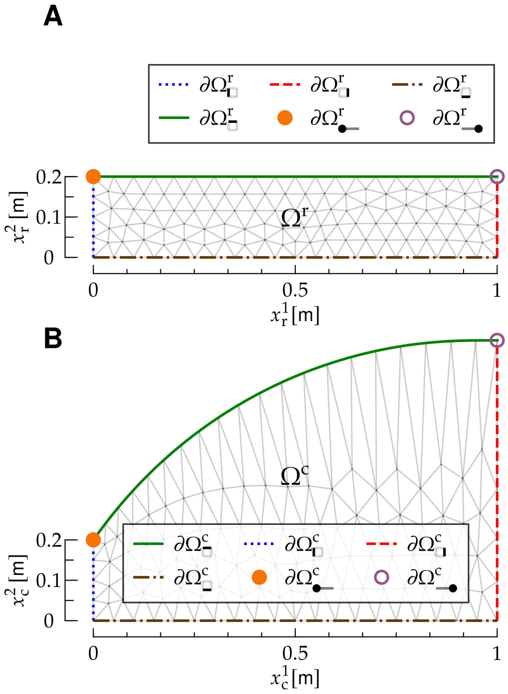

FIG. 5: Reference and current configuration for the interaction between a bulk fluid (F) and a Helfrich membrane (M). The Figure follows the same notation as Fig. 2, and it represents a vertical channel with mirror symmetry about its mid axis, which coincides with $\partial\Omega_{\square}$; the right, mirrored part of the channel is not shown. The top edge of the bulk fluid mesh, $\partial\Omega^{\mathrm{r}}_{\square}$, coincides with the one-dimensional manifold $\Omega^{\mathrm{r}}_{\_}$ of Helfrich membrane.

We enforce the following boundary conditions for ????:

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}} = \boldsymbol{v}_{\mathrm{F}\,\square} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{54}$$
$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{55}$$
$$v_{\mathrm{F}}{}^1_{\mathrm{c}} = 0 \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{56}$$
$$\frac{\partial v_{\mathrm{F}}{}^2_{\mathrm{c}}}{\partial \mathrm{x}^2_{\mathrm{c}}} = 0 \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{57}$$
$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = \left.\frac{\mathrm{d}\boldsymbol{u}_{\mathrm{M}}(x^1)}{\mathrm{d}t}\right|_{x^1 = [\boldsymbol{\psi}^t(\mathbf{x}_{\mathrm{c}})]^1} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{58}$$
$$\varsigma^{\mathrm{c}}_{\mathrm{F}} = \varsigma_{\mathrm{F}\,\square} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}. \tag{59}$$

According to **??**, bulk fluid is injected from the bottom edge of Ω^{c}, $\partial\Omega^{\mathrm{c}}_{\square}$, while **??** is a no-slip boundary condition on the left wall $\partial\Omega^{\mathrm{c}}_{\square}$, and **????** result from symmetry. **??** ensures that the motion of bulk fluid is conforming to that of Helfrich membrane [**?**]: In fact, Eq. (23) ensures that Helfrich membrane only moves normally to its profile, and **??** thus ensures that bulk fluid stays in contact with Helfrich membrane throughout its motion. On the

FIG. 6: Interaction between a bulk fluid (F) and a Helfrich membrane (M). Here, bulk fluid is air, and Helfrich membrane an elastomer, and the bulk fluid Reynolds number is $R \sim 200$. **A**) Reference (r) and current (c) configuration of Helfrich membrane (red and green curve, respectively), and Helfrich membrane displacement field $\boldsymbol{u}_{\mathrm{M}}$ (black arrows). **B**) Stretch field of Helfrich membrane. **C**) Tangent angle of Helfrich membrane. **D**) and **E**): bulk fluid velocity and tension, respectively; the notation is the same as in Fig. 1. **F**), **G**) and **H**): Tangential velocity, normal velocity and tension of Helfrich membrane, respectively. All panels refer to the same instant of time, shown on top, and they display also the deformed mesh (gray lines).

other hand, **??** implies a tangential slip between Helfrich membrane and bulk fluid, allowing for the tangential part of the bulk fluid and Helfrich membrane velocities to differ. Finally, **??** stems from the presence of the reservoir, in which the bulk fluid tension is equal to $\varsigma_{\mathrm{F}\,\square}$: At the boundary $\partial\Omega^{\mathrm{c}}_{\square}$ between the channel and the reservoir, the bulk fluid tension matches the reservoir's.

b. Equations of motion in reference configuration coordinates We will now rewrite the equations of motion and the boundary conditions of **??** in terms of $\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}$, $\varsigma^{\mathrm{r}}_{\mathrm{F}}$, and the reference coordinate $\mathbf{x}_{\mathrm{r}}$. The equations of motion **????** yield, respectively,

$$\rho_{\mathrm{F}} \left\{ \frac{\partial {v_{\mathrm{F}}}^{\alpha}_{\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t)}{\partial t} + \left[{v_{\mathrm{F}}}^{\gamma}_{\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t) - \frac{\mathrm{d}\varphi^{\gamma\, t}(\mathbf{x}_{\mathrm{r}})}{\mathrm{d}t} \right] G_{\beta\gamma}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}})) \frac{\partial {v_{\mathrm{F}}}^{\alpha}_{\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t)}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} \right\} = \tag{60}$$

$$G_{\beta\alpha}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}})) \frac{\partial \varsigma^{\mathrm{r}}_{\mathrm{F}}(\mathbf{x}_{\mathrm{r}}, t)}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} + \eta_{\mathrm{F}}\, G_{\delta\beta}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}})) \frac{\partial}{\partial \mathsf{x}^{\delta}_{\mathrm{r}}} \left[G_{\gamma\beta}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}})) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t)}{\partial \mathsf{x}^{\gamma}_{\mathrm{r}}} \right],$$

$$G_{\beta\alpha}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}})) \frac{\partial {v_{\mathrm{F}}}^{\alpha}_{\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t)}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} = 0, \tag{61}$$

which hold in Ω^{r}. In **????** the tensor G is given by the inverse of F [**?**]:

$$G_{\alpha\beta}(\boldsymbol{u}) \equiv [F(\boldsymbol{u})]^{-1}_{\alpha\beta}. \tag{62}$$

4. Temporal discretization

To numerically solve for the dynamics in a time span $0 \leq t \leq T$, we discretize time by setting $\Delta t \equiv T/N$, $t_n \equiv n\,\Delta t$, with $n = 0, 1, \cdots, N$.

We combine the Crank Nicolson discretization method for Navier-Stokes equations [**? ?**] with the arbitrary Lagrangian-Eulerian formulation, by discretizing the dynamical equations for Helfrich membrane, bulk fluid domain and bulk fluid. The dynamics of the system is obtained as follows.

Following the Crank Nicolson temporal-discretization scheme [**?**], we define fields at staggered, integer and semi-integer time steps [**?**]: For example, we set

$$ {\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{\,n}(\mathbf{x}_{\mathrm{r}}) \equiv \boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t_n), \tag{63}$$

$$ \varsigma^{\mathrm{r}\,n-1/2}_{\mathrm{F}}(\mathbf{x}_{\mathrm{r}}) \equiv \varsigma^{\mathrm{r}}_{\mathrm{F}}\left(\mathbf{x}_{\mathrm{r}}, \frac{t_n + t_{n-1}}{2}\right), \tag{64}$$

and similarly for other bulk fluid and Helfrich membrane fields.

a. Helfrich membrane We discretize Eqs. (20) to (25) as follows

$$\nabla^{n-1/2}_i v^i_{\mathrm{M}} - 2H^{n-1/2} w^n_{\mathrm{M}} = 0, \tag{65}$$

$$\rho_{\mathrm{M}}\Big(\frac{v_{\mathrm{M}}^{n,\,i}-v_{\mathrm{M}}^{n-1,\,i}}{\Delta t}+\frac{3}{2}v_{\mathrm{M}}^{n-1,\,j}\nabla_j^{n-1/2}v_{\mathrm{M}}^{n-1,\,i}-\frac{1}{2}v_{\mathrm{M}}^{n-2,\,j}\nabla_j^{n-1/2}v_{\mathrm{M}}^{n-2,\,i}-2v_{\mathrm{M}}^{n-1,\,j}w_{\mathrm{M}}^{n-1}b^{n-1/2,\,i}{}_{j}-$$
$$w_{\mathrm{M}}^{n-1}\nabla^{n-1/2,\,i}w_{\mathrm{M}}^{n-1}\Big)=$$
$$\nabla^{n-1/2,\,i}\varsigma_{\mathrm{M}}{}^{n-1/2}+f^{i}_{\eta_{\mathrm{M}}\,\mathrm{t}}\left(\frac{\boldsymbol{v}_{\mathrm{M}}{}^{n}+\boldsymbol{v}_{\mathrm{M}}{}^{n-1}}{2},\frac{w_{\mathrm{M}}^{n}+w_{\mathrm{M}}^{n-1}}{2},\nu^{n-1/2},\psi^{n-1/2}\right)+$$
$$f^{i}_{\mathrm{F\,t}}(\boldsymbol{\sigma}_{\mathrm{F}}^{\mathrm{r}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1},\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-3/2},\boldsymbol{u}_{\mathrm{D}}^{n-1}),\nu^{n-1/2},\psi^{n-1/2}), \tag{66}$$

$$\rho_{\mathrm{M}}\left(\frac{w_{\mathrm{M}}^{n}-w_{\mathrm{M}}^{n-1}}{\Delta t}+v_{\mathrm{M}}^{n-1,\,i}v_{\mathrm{M}}^{n-1,\,j}b_{ji}^{n-1/2}+\frac{3}{2}v_{\mathrm{M}}^{n-1,\,i}\nabla_i^{n-1/2}w_{\mathrm{M}}^{n-1}-\frac{1}{2}v_{\mathrm{M}}^{n-2,\,i}\nabla_i^{n-1/2}w_{\mathrm{M}}^{n-2}\right)= \tag{67}$$
$$f_{\kappa}(\nu^{n-1/2},\psi^{n-1/2})+2\varsigma_{\mathrm{M}}{}^{n-1/2}H^{n-1/2}+f_{\eta_{\mathrm{M}}\,\mathrm{n}}\left(\frac{\boldsymbol{v}_{\mathrm{M}}{}^{n}+\boldsymbol{v}_{\mathrm{M}}{}^{n-1}}{2},\frac{w_{\mathrm{M}}^{n}+w_{\mathrm{M}}^{n-1}}{2},\nu^{n-1/2},\psi^{n-1/2}\right)+$$
$$f_{\mathrm{F\,n}}(\boldsymbol{\sigma}_{\mathrm{F}}^{\mathrm{r}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1},\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-3/2},\boldsymbol{u}_{\mathrm{D}}^{n-1}),\nu^{n-1/2},\psi^{n-1/2}),$$

$$\frac{\boldsymbol{u}_{\mathrm{M}}^{n-1/2}-\boldsymbol{u}_{\mathrm{M}}^{n-3/2}}{\Delta t}=w_{\mathrm{M}}^{n-1}\hat{n}^{\mathrm{c}\,n-1/2}, \tag{68}$$
$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,1}}{\partial x^1}=-1+\nu^{n-1/2}\cos\psi^{n-1/2}, \tag{69}$$
$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,2}}{\partial x^1}=-\nu^{n-1/2}\sin\psi^{n-1/2}. \tag{70}$$

In **????**, $\sigma_{\mathrm{F}}^{\mathrm{r}}$ denotes the bulk fluid stress tensor in the reference configuration:

$$\sigma^{\mathrm{r}\,t}_{\mathrm{F}\,\alpha\beta}(\mathsf{x}_{\mathrm{r}})\equiv\sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta}(\boldsymbol{\varphi}^{t}(\mathsf{x}_{\mathrm{r}})) \tag{71}$$
$$=\varsigma_{\mathrm{F}}^{\mathrm{r}}(\mathsf{x}_{\mathrm{r}},t)\delta_{\alpha\beta}+\eta_{\mathrm{F}}\Big[G_{\gamma\beta}(\boldsymbol{u}_{\mathrm{D}}^{t}(\mathsf{x}_{\mathrm{r}}))\frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}+$$
$$G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{t}(\mathsf{x}_{\mathrm{r}}))\frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{\beta}}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}\Big].$$

Proceeding along the same lines, the discrete form of the boundary conditions **????????????????** reads, respectively,

$$v_{\mathrm{M}}^{n,\,i}=0\text{ on }\partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{72}$$
$$v_{\mathrm{M}}^{n,\,i}=0\text{ on }\partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{73}$$
$$w_{\mathrm{M}}^{n}=0\text{ on }\partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{74}$$
$$\nabla_i^{n-1/2}w_{\mathrm{M}}^{n}=0\text{ on }\partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{75}$$
$$\varsigma_{\mathrm{M}}{}^{n-1/2}=\varsigma_{\mathrm{M}\,\bullet\!-}\text{ on }\partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{76}$$
$$\boldsymbol{u}_{\mathrm{M}}^{n-1/2}=0\text{ on }\partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{77}$$
$$u_{\mathrm{M}}^{n-1/2,\,1}=0\text{ on }\partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{78}$$
$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,2}}{\partial x^1}=0\text{ on }\partial\Omega^{\mathrm{r}}_{-\!\bullet}. \tag{79}$$

In what follows, we will split the boundary-value problem into three steps [? ?]:

1. **Approximated velocities.** Proceeding along the lines of the splitting scheme for bulk fluid of **??**, we introduce the auxiliary velocities $\overline{\boldsymbol{v}}_{\mathrm{M}}$ and $\overline{w}_{\mathrm{M}}$, which approximate, respectively, $\boldsymbol{v}_{\mathrm{M}}{}^{n}$ and w_{M}^{n} [?]. We define $\overline{\boldsymbol{v}}_{\mathrm{M}}$ and $\overline{w}_{\mathrm{M}}$ as solutions of the partial differential equations

$$\rho_{\mathrm{M}}\Big[\frac{\overline{v}_{\mathrm{M}}^{i}-v_{\mathrm{M}}^{n-1,\,i}}{\Delta t}+\left(\frac{3}{2}v_{\mathrm{M}}^{n-1,\,j}-\frac{1}{2}v_{\mathrm{M}}^{n-2,\,j}\right)\nabla_j^{n-1/2}V_{\mathrm{M}}^{i}-2V_{\mathrm{M}}^{j}W_{\mathrm{M}}b^{n-1/2,\,i}{}_{j}-W_{\mathrm{M}}\nabla^{n-1/2,\,i}W_{\mathrm{M}}\Big]= \tag{80}$$
$$\nabla^{n-1/2,\,i}\varsigma_{\mathrm{M}}^{*}+f^{i}_{\eta_{\mathrm{M}},\mathrm{t}}\left(\boldsymbol{V}_{\mathrm{M}},W_{\mathrm{M}},\nu^{n-1/2},\psi^{n-1/2}\right)+$$
$$f^{i}_{\mathrm{F\,t}}(\boldsymbol{\varsigma}_{\mathrm{F}}^{\mathrm{r}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1},\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-3/2},\boldsymbol{u}_{\mathrm{D}}^{n-1}),\nu^{n-1/2},\psi^{n-1/2}),$$

$$\rho_{\mathrm{M}}\left[\frac{\overline{w}_{\mathrm{M}}-w_{\mathrm{M}}^{n-1}}{\Delta t}+V_{\mathrm{M}}^{i}V_{\mathrm{M}}^{j}b_{ji}^{n-1/2}+\left(\frac{3}{2}v_{\mathrm{M}}^{n-1,\,i}-\frac{1}{2}v_{\mathrm{M}}^{n-3/2,\,i}\right)\nabla_i^{n-1/2}W_{\mathrm{M}}\right]= \tag{81}$$
$$f_{\kappa}(\nu^{n-1/2},\psi^{n-1/2})+2\varsigma_{\mathrm{M}}^{*}H^{n-1/2}+f_{\eta_{\mathrm{M}}\,\mathrm{n}}(\boldsymbol{V}_{\mathrm{M}},W_{\mathrm{M}},\nu^{n-1/2},\psi^{n-1/2})+$$

$$f_{\mathrm{F\,n}}(\boldsymbol{\varsigma}_{\mathrm{F}}^{\mathrm{r}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1},\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-3/2},\boldsymbol{u}_{\mathrm{D}}^{n-1}),\nu^{n-1/2},\psi^{n-1/2}),$$

where we have set

$$V^{i}_{\mathrm{M}} \equiv \frac{\overline{v}^{i}_{\mathrm{M}} + {v_{\mathrm{M}}}^{n-1,\,i}}{2}, \tag{82}$$

$$W_{\mathrm{M}} \equiv \frac{\overline{w}_{\mathrm{M}} + w^{n-1}_{\mathrm{M}}}{2}, \tag{83}$$

$$\varsigma^{*}_{\mathrm{M}} \equiv {\varsigma_{\mathrm{M}}}^{n-3/2}, \tag{84}$$

???? are the analog of **????**, respectively, with a fixed pressure field $\varsigma^{*}_{\mathrm{M}}$ from the preceding step—in what follows, we will synonymously use the terms ‘pressure’ and ‘tension’ for conciseness [?].

Overall, **????** determine $\overline{v}_{\mathrm{M}}$ and $\overline{w}_{\mathrm{M}}$.

2. **Pressure correction.** Subtracting **????** and neglecting $\mathcal{O}(\Delta t)$ [?], we obtain

$$\rho_{\mathrm{M}} \frac{{v_{\mathrm{M}}}^{n,\,i} - \overline{v}^{i}_{\mathrm{M}}}{\Delta t} = -\nabla^{n-1/2,\,i} \phi_{\mathrm{M}}, \tag{85}$$

where we have defined the pressure increment as

$$\phi_{\mathrm{M}} \equiv \varsigma^{*}_{\mathrm{M}} - {\varsigma_{\mathrm{M}}}^{n-1/2}. \tag{86}$$

Proceeding along the same lines for **????**, we obtain

$$w^{n}_{\mathrm{M}} = \overline{w}_{\mathrm{M}}. \tag{87}$$

By taking the covariant derivative of **??** and using the mass-conservation equation **??** and **??**, we obtain

$$\nabla^{n-1/2}_{i} \nabla^{n-1/2,\,i} \phi_{\mathrm{M}} = \frac{\rho_{\mathrm{M}}}{\Delta t} (\nabla^{n-1/2}_{i} \overline{v}^{i}_{\mathrm{M}} - 2 H^{n-1/2} \overline{w}_{\mathrm{M}}). \tag{88}$$

?? is a second-order partial differential equation which determines ϕ_{M}, and thus the tension ${\varsigma_{\mathrm{M}}}^{n-1/2}$ through **??**.

3. **Velocity correction.** The Helfrich membrane velocity ${\boldsymbol{v}_{\mathrm{M}}}^{n}$ is determined by means of **??** in terms of $\overline{\boldsymbol{v}}_{\mathrm{M}}$ and ϕ_{M}.

The boundary conditions for **????????????????** are

$$\overline{v}^{i}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{89}$$

$$\overline{v}^{i}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{90}$$

$$\overline{w}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{91}$$

$$\nabla^{n-1/2}_{i} \overline{w}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{92}$$

$$\phi_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{93}$$

$${n_{\mathrm{M}}}^{n-1/2}_{i} \nabla^{n-1/2,\,i} \phi_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-}, \tag{94}$$

$${\boldsymbol{u}_{\mathrm{M}}}^{n-1/2} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{95}$$

$${u_{\mathrm{M}}}^{n-1/2,\,1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{96}$$

$$\frac{\partial {u_{\mathrm{M}}}^{n-1/2,\,2}}{\partial x^{1}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}. \tag{97}$$

In **??**, $\boldsymbol{n}_{\mathrm{M}}$ is the unit normal to $\partial\Omega_{-}$: It lies in the tangent bundle of Ω_{-}, it is directed outside Ω_{-}, and it is normalized according to ${n_{\mathrm{M}}}^{i} {n_{\mathrm{M}}}_{i} = 1$ [? ? ?].

b. Bulk fluid domain The discretization of the equations of motion for bulk fluid domain proceeds along the same lines as in Section II A, see **??**. The discrete version of the partial differential equations **????** is

$$\frac{\partial S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}^{n}_{\mathrm{D}})}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{98}$$

$$\frac{\partial}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} \frac{\mathrm{d} S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}^{n}_{\mathrm{D}})}{\mathrm{d}t} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{99}$$

with boundary conditions given by the discrete version of **????????????????**:

$$\boldsymbol{u}^{n}_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\sqcap} \cup \partial\Omega^{\mathrm{r}}_{\sqcup}, \tag{100}$$

$${u_{\mathrm{D}}}^{n,\,1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsupset}, \tag{101}$$

$$\frac{\partial u^{n,\,2}_{\mathrm{D}}}{\partial \mathsf{x}^{1}_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsupset}, \tag{102}$$

$$\boldsymbol{u}^{n}_{\mathrm{D}} = {\boldsymbol{u}_{\mathrm{M}}}^{n-1/2} \text{ on } \partial\Omega^{\mathrm{r}}_{\sqcup}, \tag{103}$$

$$\dot{\boldsymbol{u}}^{n}_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\sqcap} \cup \partial\Omega^{\mathrm{r}}_{\sqcup}, \tag{104}$$

$${\dot{u}_{\mathrm{D}}}^{n,\,1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsupset}, \tag{105}$$

$$\frac{\partial \dot{u}^{n,\,2}_{\mathrm{D}}}{\partial \mathsf{x}^{1}_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsupset}, \tag{106}$$

$$\dot{\boldsymbol{u}}^{n}_{\mathrm{D}} = {\boldsymbol{u}_{\mathrm{M}}}^{n-1/2} \text{ on } \partial\Omega^{\mathrm{r}}_{\sqcup}. \tag{107}$$

c. Bulk fluid The discretization of the equations of motion for bulk fluid proceeds along the same lines as in Section II A, and is split into three steps: Approximated velocity, pressure correction and velocity correction, see **??**.

The resulting partial differential equations for $\overline{\boldsymbol{v}}_{\mathrm{F}}$ and $\varsigma^{\mathrm{r}}_{\mathrm{F}}$ are

$$\rho_{\mathrm{F}}\left\{\frac{\overline{v}_{\mathrm{F}}^{\alpha}-{v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\Delta t}+\left[\frac{3}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\gamma}-\dot{u}_{\mathrm{D}}^{n-1,\,\gamma})\,G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-1})-\frac{1}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\gamma}-\dot{u}_{\mathrm{D}}^{n-2,\,\gamma})\,G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-2})\right]\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\right\}= \tag{108}$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial \varsigma_{\mathrm{F}}^{\mathrm{r}\,*}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}+\eta_{\mathrm{F}}\,G_{\delta\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\delta}}\left(G_{\gamma\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}\right),$$

$$\frac{\rho_{\mathrm{F}}}{\Delta t}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial \overline{v}_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}= \tag{109}$$

$$G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}\left(G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial \phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\right),$$

where we have set

$$\boldsymbol{V}_{\mathrm{F}}\equiv\frac{\overline{\boldsymbol{v}}_{\mathrm{F}}+{\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n-1}}{2}, \tag{110}$$

$$\varsigma_{\mathrm{F}}^{\mathrm{r}\,*}\equiv\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-3/2}, \tag{111}$$

$$\phi_{\mathrm{F}}\equiv\varsigma_{\mathrm{F}}^{\mathrm{r}\,*}-\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-1/2}. \tag{112}$$

The boundary conditions for **????** are [**?**]

$$\overline{\boldsymbol{v}}_{\mathrm{F}}=\boldsymbol{v}_{\mathrm{F}\,\sqcup}\text{ on }\partial\Omega_{\sqcup}^{\mathrm{r}}, \tag{113}$$

$$\overline{\boldsymbol{v}}_{\mathrm{F}}=\boldsymbol{0}\text{ on }\partial\Omega_{\sqsubset}^{\mathrm{r}}, \tag{114}$$

$$\overline{v}_{\mathrm{F}}^{1}=0\quad\text{on }\partial\Omega_{\sqsupset}^{\mathrm{r}}, \tag{115}$$

$$G(\boldsymbol{u}_{\mathrm{D}}^{n-1})_{\alpha 1}\,\frac{\partial V_{\mathrm{F}}^{2}}{\partial \mathsf{x}_{\mathrm{r}}^{\alpha}}=0\quad\text{on }\partial\Omega_{\sqsupset}^{\mathrm{r}}, \tag{116}$$

$$\overline{\boldsymbol{v}}_{\mathrm{F}}(x^{1},h)=\dot{u}_{\mathrm{M}}^{n-1/2}(x^{1})\text{ on }\partial\Omega_{\sqcap}^{\mathrm{r}}, \tag{117}$$

$$\phi_{\mathrm{F}}=0\text{ on }\partial\Omega_{\sqcup}^{\mathrm{r}}, \tag{118}$$

$$\hat{n}^{\mathrm{r}\,\gamma}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial \phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}= \tag{119}$$

$$0\text{ on }\partial\Omega_{\sqsubset}^{\mathrm{r}}\cup\partial\Omega_{\sqcap}^{\mathrm{r}},$$

$$G(\boldsymbol{u}_{\mathrm{D}}^{n-1})_{\alpha 1}\,\frac{\partial \phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\alpha}}=0\text{ on }\partial\Omega_{\sqsupset}^{\mathrm{r}}. \tag{120}$$

Finally, ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n}$ is determined from $\overline{\boldsymbol{v}}_{\mathrm{F}}$ and $\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-1/2}$ by means of the relation

$$\rho_{\mathrm{F}}\frac{\overline{v}_{\mathrm{F}}^{\alpha}-{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}}{\Delta t}=G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial \phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\text{ in }\Omega^{\mathrm{r}}. \tag{121}$$

Given the Helfrich membrane fields $v_{\mathrm{M}}^{n-1,i}$, $w_{\mathrm{M}}^{n-1,\,i}$, ${\varsigma_{\mathrm{M}}}^{n-3/2}$, $\boldsymbol{u}_{\mathrm{M}}^{n-3/2}$, $\nu^{n-3/2}$, $\psi^{n-3/2}$, the bulk fluid domain fields $\boldsymbol{u}_{\mathrm{D}}^{n-1}$, $\dot{\boldsymbol{u}}_{\mathrm{D}}^{n-1}$, and the bulk fluid fields ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n-1}$, $\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-3/2}$, we determine the fields at the next time step $n\to n+1$ as follows:

- **Update Helfrich membrane.** The Helfrich membrane fields $v_{\mathrm{M}}^{n,i}$, $w_{\mathrm{M}}^{n,\,i}$, ${\varsigma_{\mathrm{M}}}^{n-1/2}$, $\boldsymbol{u}_{\mathrm{M}}^{n-1/2}$, $\nu^{n-1/2}$, $\psi^{n-1/2}$ are determined by solving the boundary-value problem given by **?????????????????????????????????**.
- **Update bulk fluid domain.** We determine the bulk fluid domain displacement field and its time derivative, $\boldsymbol{u}_{\mathrm{D}}^{n}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}^{n}$, from the boundary-value problems given by **????????????????????**.
- **Update bulk fluid.** We determine the bulk fluid fields ${\boldsymbol{v}_{\mathrm{M}}}^{n}$ and $\varsigma_{\mathrm{F}}^{\mathrm{r}\,n-1/2}$ by solving the boundary-value problem given by **????????????????????**, and by solving **??**.

The fields ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{c}}^{n}$, $\varsigma_{\mathrm{F}}^{\mathrm{c}\,n-1/2}$ in the current configuration are then obtained from the respective fields in the reference configuration through Eqs. (5) and (6).

An example of this dynamics is shown in **??**, where we show the interaction between air (bulk fluid) and an elastomer (Helfrich membrane). The channel has dimensions $L=1\,\mathrm{m}$, $h=0.2\,\mathrm{m}$. Proceeding along the lines of Section II A, Helfrich membrane parameters have been estimated by considering the parameters for a three-dimensional material, multiplying them by a unit length corresponding to the out-of-plane depth, and by the Helfrich membrane thickness, chosen to be $\Delta=10^{-3}\,\mathrm{m}$.

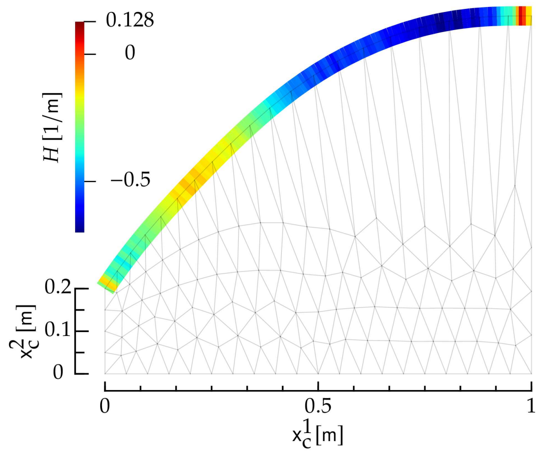

FIG. 7: Curvature of the Helfrich membrane interacting with a bulk fluid shown in **??**. The Figure follows the same notation and it corresponds to the same instant of time as **??**.

The Helfrich membrane bending rigidity κ has been estimated from the relation $\kappa = E\Delta^3/[12(1-\nu^2)]$, which expresses κ as a function of the Young modulus E, the Poisson ratio ν, and Δ [**?**]. Typical values for an elastomer are $E \sim 10^7\,\mathrm{Pa}$ [**?**] and $\nu \sim 0.5$ [**?**], yielding $\kappa = 10^{-3}\,\mathrm{Kg\,m^3/\,sec^2}$. We chose $\rho_\mathrm{M} = 1\,\mathrm{Kg/m}$ and $\eta_\mathrm{M} = 1\,\mathrm{Kg\,m/\,sec}$ as the density and viscosity of an elastomer, respectively [**?** **?**], and we take $\varsigma_{\mathrm{M}\bullet\!-} = 1\,\mathrm{N}$. The bulk fluid parameters have been estimated from the air parameters: $\rho_\mathrm{F} = 1\,\mathrm{Kg/m^2}$, $\eta_\mathrm{F} = 10^{-5}\,\mathrm{Kg/\,sec}$ [**?**]. Also, we take $\varsigma_{\mathrm{F}\,\square} = -1\,\mathrm{N/m^2}$, and $\boldsymbol{v}^1_{\mathrm{F}\,\square} = 0$, $\boldsymbol{v}^2_{\mathrm{F}\,\square}(\mathbf{x}_\mathrm{c}) = \frac{3}{2L^2}\mathsf{x}^1_\mathrm{c}(2L-\mathsf{x}^1_\mathrm{c})v_0$, where $v_0 = 10^{-2}\,\mathrm{m/\,sec}$, and $\boldsymbol{v}^2_{\mathrm{F}\,\square}$ corresponds to a Poiseuille flow [**?**]. Finally, the bulk fluid domain exponent is $\zeta = 3$ [**?**].

III. DISCUSSION

The Helfrich membrane shape shown in **??** has a curvature which varies along the Helfrich membrane profile, and thus differs from the classical circle-arc obtained at the interface between two fluids [**?** **?**].

ACKNOWLEDGMENTS

We would like to thank S. Al-Izzi and S. Astruc for useful discussions, and the finite element computational software online community for its precious support.

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Appendix A: Fluid and rigid body

1. Rigid body motion

In this Section, we will work out the equations of motion for rigid body.

a. *Equation of motion in current-configuration coordinates*

The dynamics of rigid body is given by the equations of motion for a rigid body which, in current coordinates, read [?]

$$I\frac{\mathrm{d}\omega}{\mathrm{d}t} = -\int_{\partial\Omega^{\mathrm{c}}_{\circ}} \mathrm{d}l_{\mathrm{c}}\, \epsilon_{3\alpha\beta}\, \sigma^{\mathrm{c}}_{\mathrm{F}\,\beta\gamma} \hat{n}^{\mathrm{c}\,\gamma}, \tag{A1}$$

$$\omega \equiv \frac{\mathrm{d}\theta}{\mathrm{d}t}. \tag{A2}$$

?? relates the angular acceleration of rigid body to the torque of external forces. The moment of inertia of rigid body is denoted by I. The right-hand side of **??** contains the line integral along the boundary of rigid body, of the torque of the local force exerted by bulk fluid on rigid body with respect to the left focal point $\boldsymbol{f}$ of rigid body [?], and $\hat{n}^{\mathrm{c}}$ is the normal to $\partial\Omega^{\mathrm{c}}_{\circ}$ pointing towards Ω^{c}.

This force is expressed in terms of the bulk fluid stress tensor [?] in current coordinates $\sigma^{\mathrm{c}}_{\mathrm{F}}$, given by Eq. (17), where η_{F} and ρ_{F} are the two-dimensional viscosity and density of bulk fluid, respectively. Also, $\epsilon_{\alpha\beta\gamma}$ is the three-dimensional Levi-Civita symbol. Finally, $\mathrm{d}l_{\mathrm{c}}$ is the line element of $\partial\Omega^{\mathrm{c}}_{\circ}$ in the current configuration, and $\hat{\boldsymbol{n}}^{\mathrm{c}}$ the unit boundary normal to $\partial\Omega^{\mathrm{c}}_{\circ}$ pointing outside Ω^{c}.

b. *Equation of motion in the reference-configuration coordinates*

In reference coordinates, **??** reads

$$I\frac{\mathrm{d}\omega}{\mathrm{d}t} = \int_{\partial\Omega^{\mathrm{r}}_{\circ}} \mathrm{d}s\, \epsilon_{\alpha\beta}\left[\varphi^{\alpha,\,t}(\boldsymbol{\xi}(s)) - f^{\alpha}\right] \sigma^{\mathrm{r}}_{\mathrm{F}\,\beta\gamma}(\boldsymbol{\xi}(s)) \epsilon_{\gamma\lambda} F_{\lambda\mu}(\boldsymbol{u}^{t}_{\mathrm{D}}(\boldsymbol{\xi}(s)))\frac{\mathrm{d}\xi^{\mu}(s)}{\mathrm{d}s}. \tag{A3}$$

In **??**, the integral in the right-hand side is over the curvilinear boundary $\partial\Omega^{\mathrm{r}}_{\circ}$, which is parametrized as $\mathbf{x}_{\mathrm{r}} = \boldsymbol{\xi}(s)$, where s is the curvilinear coordinate. Also, $\epsilon_{\alpha\beta}$ is the two-dimensional Levi-Civita symbol [?], and $\sigma^{\mathrm{r}}_{\mathrm{F}}$ denotes the stress tensor in the reference configuration, which is given by **??**, where G is given by **??**.

2. Fluid motion

In this Section, we will work out the equations of motion for bulk fluid.

a. *Equations of motion in the current-configuration coordinates*

The equations of motion, expressed in terms of the current fields $v_{\mathrm{F}\,\mathrm{c}}$ and $\varsigma^{\mathrm{c}}_{\mathrm{F}}$, are **????**, see **??** for details.

The boundary conditions for **????????** are

$$v_{\mathrm{F}}{}^{1}_{\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = \boldsymbol{v}_{\mathrm{F}\,\sqsubset} \text{ on } \partial\Omega^{\mathrm{c}}_{\sqsubset}, \tag{A4}$$

$$v_{\mathrm{F}}{}^{2}_{\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = 0 \text{ on } \partial\Omega^{\mathrm{c}}_{\sqcap}, \tag{A5}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{c}}_{\sqcup}, \tag{A6}$$

$$v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = \frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}[\psi^{t\,\beta}(\mathbf{x}_{\mathrm{c}}) - f^{\beta}] \text{ on } \partial\Omega^{\mathrm{c}}_{\circ}, \tag{A7}$$

$$\eta_{\mathrm{F}}\frac{\partial v_{\mathrm{F}}{}^{\alpha}_{\mathrm{c}}}{\partial \mathrm{x}^{1}_{\mathrm{c}}} = 0 \text{ on } \partial\Omega^{\mathrm{c}}_{\sqsupset}, \tag{A8}$$

$$\varsigma^{\mathrm{c}}_{\mathrm{F}}(\mathbf{x}_{\mathrm{c}}) = 0 \text{ on } \partial\Omega^{\mathrm{c}}_{\sqsupset}. \tag{A9}$$

???? ensure that bulk fluid is injected on the boundary $\partial\Omega^{\mathrm{c}}_{\sqsubset}$ in a direction parallel to the $\mathsf{x}^1_{\mathrm{c}}$ axis, and **??** is a no-slip boundary condition [**?**] along the channel walls $\partial\Omega^{\mathrm{c}}_{\sqcup}$, see Fig. 2. **??** ensures that, at the boundary $\partial\Omega^{\mathrm{c}}_{\circ}$, the bulk fluid velocity matches the rotational velocity of rigid body. Given that rigid body can only rotate about its left focal point $\boldsymbol{f}$, the rigid body rotational velocity of a point $\mathbf{x}_{\mathrm{c}} \in \partial\Omega^{\mathrm{c}}_{\circ}$ is given by

$$\frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}[\psi^{t\,\beta}(\mathbf{x}_{\mathrm{c}}) - c_\beta], \tag{A10}$$

where $\boldsymbol{R}$ is the rotation matrix corresponding to a counterclockwise rotation:

$$\boldsymbol{R}(\theta) \equiv \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{A11}$$

Finally, **????** ensure, respectively, that on $\partial\Omega^{\mathrm{c}}_{\sqsupset}$ there is zero traction and tension; this corresponds to the hypothesis that there is free flow at $\partial\Omega^{\mathrm{c}}_{\sqsupset}$ [**?** **?**].

b. Equations of motion in the reference-configuration coordinates

We will now rewrite the equations of motion and the boundary conditions in terms of $\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}$, $\varsigma^{\mathrm{r}}_{\mathrm{F}}$, and the reference coordinate $\mathbf{x}_{\mathrm{r}}$. The equations of motion **????** yield, respectively, **????**, see **??**. The boundary conditions **????????????** yield as

$$v_{\mathrm{F}\,\mathrm{r}}^{\,1}(\mathbf{x}_{\mathrm{r}}) = \boldsymbol{v}_{\mathrm{F}\,\sqsubset} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsubset}, \tag{A12}$$

$$v_{\mathrm{F}\,\mathrm{r}}^{\,2}(\mathbf{x}_{\mathrm{r}}) = 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsubset}, \tag{A13}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}(\mathbf{x}_{\mathrm{r}}) = \mathbf{0} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqcup}, \tag{A14}$$

$$v_{\mathrm{F}\,\mathrm{r}}^{\,\alpha}(\mathbf{x}_{\mathrm{r}}) = \frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}(\mathsf{x}^\beta_{\mathrm{r}} - f^\beta) \ \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{A15}$$

$$\eta_{\mathrm{F}} G_{\beta 1}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}}))\frac{\partial v_{\mathrm{F}\,\mathrm{r}}^{\,\alpha}}{\partial \mathsf{x}^\beta_{\mathrm{r}}} = 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsupset}, \tag{A16}$$

$$\varsigma^{\mathrm{r}}_{\mathrm{F}}(\mathbf{x}_{\mathrm{r}}) = 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsupset}. \tag{A17}$$

3. Motion of bulk fluid domain

The equations of motion **??????????????????** involve the displacement field $\boldsymbol{u}_{\mathrm{D}}$, which describes how the region—or domain—which contains bulk fluid, is deformed in time, cf. Fig. 2. We will denote this region as the bulk fluid domain. We observe that this field has, so far, been left arbitrary, and we have the freedom to choose it.

Here, $\boldsymbol{u}_{\mathrm{D}}$ will be determined with an analogy with the theory of elasticity [**?**]. We imagine that, as rigid body rotates about its focal point, each point $\mathbf{x}_{\mathrm{r}} \in \Omega^{\mathrm{r}}$ moves into a point $\mathbf{x}_{\mathrm{c}} \in \Omega^{\mathrm{c}}$, and that the domain Ω^{r} is deformed, e.g., locally stretched and compressed, into Ω^{c}, like an elastic medium. We will determine $\boldsymbol{u}_{\mathrm{D}}$ as the solution of a boundary-value problem given by the elastic model **??**, see **??** for details.

The boundary conditions for the partial differential equation **??** are

$$\boldsymbol{u}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{A18}$$

$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{f} - \mathbf{x}_{\mathrm{r}} + \boldsymbol{R}(\theta)\cdot(\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}. \tag{A19}$$

?? enforces the fact that the mesh is not deformed at the boundary $\partial\Omega^{\mathrm{r}}_{\square}$. Finally, **??** ensures that the mesh deformation at $\partial\Omega^{\mathrm{r}}_{\circ}$ matches the deformation induced by the rotation of rigid body.

?????? constitute the boundary-value problem which determines $\boldsymbol{u}_{\mathrm{D}}$. Along the lines of boundary-value problems in the theory of elasticity, such boundary-value problem is already, naturally formulated in the reference configuration [**?**].

From the boundary-value problem above, we obtain a boundary-value problem for $\dot{\boldsymbol{u}}_{\mathrm{D}}$, which will be needed in the following to describe the bulk fluid domain motion. By deriving both sides of **??????**, we obtain, respectively, **??** and

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{A20}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \omega\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\cdot(\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{A21}$$

where in **????** we used the fact that $\mathbf{x}_{\mathrm{r}}$ is independent of time, and in **??** we substituted **??**. The left-hand side of **??** depends on the second partial derivatives of $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$—see **??????????**. As a result, given θ, ω, and $\boldsymbol{u}_{\mathrm{D}}$ from **??????**, the boundary-value problem **??????** determines $\dot{\boldsymbol{u}}_{\mathrm{D}}$.

Overall **??????????????????????????????????** constitute a boundary-value problem for $\theta, \omega, \boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}, \varsigma^{\mathrm{r}}_{\mathrm{F}}$, $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$ which, combined with the temporal boundary conditions

$$\theta(t=0) = \theta_0 \tag{A22}$$

$$\omega(t=0) = \omega_0 \tag{A23}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}(\mathbf{x}_{\mathrm{r}}, t=0) = \boldsymbol{v}_{\mathrm{F}\,\mathrm{r}\,0}(\mathbf{x}_{\mathrm{r}}), \tag{A24}$$

$$\varsigma^{\mathrm{r}}_{\mathrm{F}}(\mathbf{x}_{\mathrm{r}}, t=0) = \varsigma^{\mathrm{r}}_{\mathrm{F}\,0}(\mathbf{x}_{\mathrm{r}}), \tag{A25}$$

determine the dynamics of the system.

4. Temporal discretization

To numerically solve for the dynamics in a time span $0 \leq t \leq T$, we discretize time by setting $\Delta t \equiv T/N$, $t_n \equiv n\,\Delta t$, with $n = 0, 1, \cdots, N$.

We combine the Crank Nicolson discretization method for Navier-Stokes equations [? ?] with the arbitrary Lagrangian-Eulerian formulation, by discretizing the dynamical equations for rigid body, bulk fluid and bulk fluid domain—see below.

a. *Rigid body*

Proceeding along the lines of the Crank Nicolson scheme [?], we define fields at staggered, integer and semi-integer time steps [?]—see for example **????** for $\boldsymbol{v}_{\mathrm{F_r}}$ and $\varsigma^{\mathrm{r}}_{\mathrm{F}}$—and similarly for other quantities. We discretize **????** and, neglecting $\mathcal{O}(\Delta t)$, we obtain

$$\frac{\theta^n - \theta^{n-1}}{\Delta t} = \omega^n, \tag{A26}$$

$$\frac{\omega^n - \omega^{n-1}}{\Delta t} = I \int_{\partial\Omega^{\mathrm{r}}_{\mathrm{O}}} \mathrm{d}s\, \epsilon_{\alpha\beta} \left[\xi^\alpha(s) + u^{n-1,\,\alpha}(\boldsymbol{\xi}(s)) - f^\alpha\right] \sigma^{\mathrm{r}\;n-1}_{\mathrm{F}\,\beta\gamma}(\boldsymbol{\xi}(s); {\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n-1}, \varsigma^{\mathrm{r}\;n-3/2}_{\mathrm{F}}) \times \epsilon_{\gamma\lambda} F_{\lambda\mu}(\boldsymbol{u}^{n-1}_{\mathrm{D}}(\boldsymbol{\xi}(s))) \frac{\mathrm{d}\xi^\mu(s)}{\mathrm{d}s}, \tag{A27}$$

where in **??** we used Eqs. (1) and (3) and we explicitly indicated the dependence of the stress tensor **??** on $\boldsymbol{v}_{\mathrm{F_c}}$ and $\varsigma^{\mathrm{c}}_{\mathrm{F}}$.

b. *Bulk fluid*

To discretize in time the bulk fluid problem, we will apply the Crank Nicolson splitting scheme [? ?] to the Navier-Stokes equations **????** in the reference configurations. This splitting scheme is derived form the incremental pressure correction scheme [?], in which the solution of the Navier-Stokes equations is split into three intermediate steps. In what follows, we will synonymously use the terms 'pressure' and 'surface tension' for conciseness [?].

The discrete form of the partial differential equations **????** is

$$\rho_{\mathrm{F}} \left[\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha} - {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\Delta t} + \frac{3}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\gamma} - \dot{u}^{n-1,\,\gamma})\, G_{\beta\gamma}(\boldsymbol{u}^{n-1}_{\mathrm{D}}) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\partial \mathrm{x}^\beta_{\mathrm{r}}} - \frac{1}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\gamma} - \dot{u}^{n-2,\,\gamma}) \times G_{\beta\gamma}(\boldsymbol{u}^{n-2}_{\mathrm{D}}) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\alpha}}{\partial \mathrm{x}^\beta_{\mathrm{r}}} \right] = G_{\beta\alpha}(\boldsymbol{u}^{n-1}_{\mathrm{D}})\, \frac{\partial \varsigma^{\mathrm{r}\;n-1/2}_{\mathrm{F}}}{\partial \mathrm{x}^\beta_{\mathrm{r}}} + \eta_{\mathrm{F}}\, G_{\delta\beta}(\boldsymbol{u}^{n-1}_{\mathrm{D}})\, \frac{\partial}{\partial \mathrm{x}^\delta_{\mathrm{r}}} \left[G_{\gamma\beta}(\boldsymbol{u}^{n-1}_{\mathrm{D}}) \frac{\partial}{\partial \mathrm{x}^\gamma_{\mathrm{r}}} \left(\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha} + {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{2} \right) \right], \tag{A28}$$

$$G_{\beta\alpha}(\boldsymbol{u}^{n-1}_{\mathrm{D}}) \frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}}{\partial \mathrm{x}^\beta_{\mathrm{r}}} = 0, \tag{A29}$$

where we omitted the dependence on $\mathbf{x}_{\mathrm{r}}$ for clarity. Also, in the left-hand side of **??** we rewrote the convective term according to the Adams-Bashforth discretization scheme [? ?], and we used Eqs. (1) and (3) to rewrite the time derivative of $\boldsymbol{\varphi}$ in terms of $\dot{\boldsymbol{u}}_{\mathrm{D}}$.

The discrete form of the boundary conditions ???????????? reads

$$
\begin{aligned}
{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,1} &= \boldsymbol{v}_{\mathrm{F}\,\square} \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A30)}\\
{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,2} &= 0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A31)}\\
{\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{\,n} &= \boldsymbol{0} \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A32)}\\
{\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}} &= \omega^n \left.\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\right|_{\theta=\theta_n} \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, && \text{(A33)}\\
\eta_{\mathrm{F}}\, G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} \left(\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha} + {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{2} \right) &= 0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A34)}\\
{\varsigma_{\mathrm{F}}^{\mathrm{r}}}^{\,n-1/2} &= 0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}. && \text{(A35)}
\end{aligned}
$$

We will now split the boundary-value problem into three steps [? ?]:

1. **Approximated velocity.** We introduce the velocity $\overline{\boldsymbol{v}}_{\mathrm{F}}$, which approximates the true velocity ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}$, and which satisfies the boundary-value problem given by **??** and boundary conditions

$$
\begin{aligned}
\overline{v}_{\mathrm{F}}^{1} &= \boldsymbol{v}_{\mathrm{F}\,\square} \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A36)}\\
\overline{v}_{\mathrm{F}}^{2} &= 0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A37)}\\
\overline{\boldsymbol{v}}_{\mathrm{F}} &= \boldsymbol{0} \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}, && \text{(A38)}\\
\overline{\boldsymbol{v}}_{\mathrm{F}} &= \omega^n \left.\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\right|_{\theta=\theta_n} \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, && \text{(A39)}\\
\eta_{\mathrm{F}}\, G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} &= 0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}. && \text{(A40)}
\end{aligned}
$$

   where $\boldsymbol{V}_{\mathrm{F}}$ and ${\varsigma_{\mathrm{F}}^{\mathrm{r}}}^{*}$ are given by **????**.

2. **Pressure correction.**

   Subtracting **????** and neglecting $\mathcal{O}(\Delta t)$, we obtain **??**, where the surface-tension increment ϕ_{F} is defined by **??**. By applying $G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\partial/\partial \mathsf{x}_{\mathrm{r}}^{\gamma}$ to both sides of **??** and using **??**, we obtain the second-order partial differential equation **??** for ϕ_{F}.

   We will now work out the boundary conditions for **??**. **??????** imply the boundary condition

$$
\phi_{\mathrm{F}} = 0 \;\text{ on } \partial\Omega^{\mathrm{r}}_{\square}. \tag{A41}
$$

   Multiplying **??** by $G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\hat{n}^{\mathrm{r}\,\gamma}$, where $\hat{\boldsymbol{n}}^{\mathrm{r}}$ is the unit boundary normal in the reference configuration, we obtain

$$
\begin{aligned}
&\frac{\rho_{\mathrm{F}}}{\Delta t} G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\hat{n}^{\mathrm{r}\,\gamma}(\overline{v}_{\mathrm{F}}^{\alpha} - {v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}) = && \text{(A42)}\\
&\hat{n}^{\mathrm{r}\,\gamma} G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial \phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}.
\end{aligned}
$$

Combining **??** with **????????????????**, we obtain the second boundary condition

$$\hat{n}^{\mathrm{r}\,\gamma} G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1}) \frac{\partial \phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square} \cup \partial\Omega^{\mathrm{r}}_{\boxminus} \cup \partial\Omega^{\mathrm{r}}_{\circ}. \tag{A43}$$

The boundary-value problem **??????** determines ϕ_{F} and, through the definition **??**, the surface tension ${\varsigma_{\mathrm{c}}}^{n-1/2}$.

3. **Velocity correction.** The velocity ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{\,n}$ is determined from the approximate velocity $\overline{\boldsymbol{v}}_{\mathrm{F}}$ from **??**.

c. *Bulk fluid domain*

The discrete version of the boundary-value problem for $\boldsymbol{u}_{\mathrm{D}}$, **??????**, is given by **??** and

$$\boldsymbol{u}_{\mathrm{D}}^{n} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\blacksquare}, \tag{A44}$$

$$\boldsymbol{u}_{\mathrm{D}}^{n} = \boldsymbol{f} - \mathbf{x}_{\mathrm{r}} + \boldsymbol{R}(\theta^{n}) \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}. \tag{A45}$$

The discrete version of the boundary-value problem for $\dot{\boldsymbol{u}}_{\mathrm{D}}$, **??????**, is given by **??** and

$$\dot{\boldsymbol{u}}_{\mathrm{D}}^{n} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\blacksquare}, \tag{A46}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}}^{n} = \omega \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta} \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}. \tag{A47}$$

We can now iterate in time for the ensemble of fields of the bulk fluid-rigid body system as follows. At each step n, given θ^{n-1}, ω^{n-1}, ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{\,n-1}$, ${\varsigma_{\mathrm{F}}^{\mathrm{r}}}^{\,n-3/2}$, $\boldsymbol{u}_{\mathrm{D}}^{n-1}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}^{n-1}$ from the preceding step, we

1. **Update rigid body.** Solve the algebraic equations **????** for θ^{n} and ω^{n}.

2. **Update bulk fluid domain.** Solve the boundary-value problems **????????????** for $\boldsymbol{u}_{\mathrm{D}}^{n}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}^{n}$.

3. **Update bulk fluid.** Solve the boundary-value problems **????????????** and **??????** for $\overline{\boldsymbol{v}}_{\mathrm{F}}$ and ${\varsigma_{\mathrm{F}}^{\mathrm{r}}}^{\,n-1/2}$, and obtain ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{\,n}$ from **??**.

The fields ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{c}}^{\,n-1}$, ${\varsigma_{\mathrm{F}}^{\mathrm{c}}}^{\,n-3/2}$ in the current configuration are then obtained from the respective fields in the reference configuration through Eqs. (5) and (6).

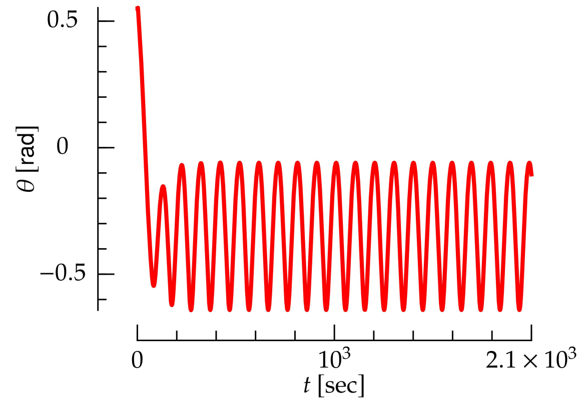

FIG. 8: Angle of the rigid body as a function of time for the interaction between a bulk fluid and a rigid body shown in Fig. 1.

5. Characteristic time scale

As shown in **??**, the dynamics of Fig. 1 involves a characteristic time scales τ_{B}, i.e., the period of the angular oscillations of body. This time scale can be estimated by considering the dynamical equation **??** for body, and approximating its left-hand side as I/τ_{B}^2 and its right-hand side as $(\eta_{\mathrm{F}} v_0 / a\, 2\pi a)a$. Here, $\eta_{\mathrm{F}} v_0/a$ is the viscous component of the force per unit length exerted by bulk fluid on body, cf. Eq. (17), $2\pi a$ estimates the total length of the boundary of body, and a approximates the lever arm of the force above. Putting everything together, we obtain

$$\tau_{\mathrm{B}} \sim \sqrt{\frac{I}{2\pi a \eta_{\mathrm{F}} v_0}}. \tag{A48}$$

For the parameters of Fig. 1, **??** yields $\tau_{\mathrm{B}} \sim 130$ sec, which is in agreement with the period of the oscillations in **??**.

Appendix B: Fluid and elastic body

1. Equations of motion

The equations of motion for the dynamics of bulk fluid and elastic body are the following.

First, the dynamics of bulk fluid is governed by the Navier-Stokes and mass-conservation equations **????**, respectively. Their boundary conditions are **??????????**, which stay unchanged with respect to Section II A. On the other hand, boundary condition ??, which ensures that the bulk fluid velocity matches the body velocity at the interface between bulk fluid and body, now reads

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_\mathrm{c}) = \left.\frac{\mathrm{d}\boldsymbol{u}_\mathrm{E}(\mathbf{x}_\mathrm{r})}{\mathrm{d}t}\right|_{\mathbf{x}_\mathrm{r}=\psi^t(\mathbf{x}_\mathrm{c})} \text{ on } \partial\Omega^\mathrm{c}_{\bigcirc} \tag{B1}$$

Second, the motion of elastic body is governed by Newton's equations of motion for an elastic body [?]

$$\rho_\mathrm{E}\frac{\mathrm{d}^2 u^\alpha_\mathrm{E}(\mathbf{x}_\mathrm{r})}{\mathrm{d}t^2} = \frac{\partial S_{\mathrm{E}\,\alpha\beta}(\boldsymbol{u}_\mathrm{E})}{\partial \mathsf{x}^\beta_\mathrm{r}}, \tag{B2}$$

where ρ_E is the density of elastic body, which is assumed to be independent of space. Also, $\boldsymbol{u}_\mathrm{E}$ is the deformation field of elastic body, which is defined along the lines of Eq. (1). The elastic body stress tensor is derived from a compressible neo-Hookean model [?], and it reads

$$S_{\mathrm{E}\,\alpha\beta} \equiv \frac{\mu}{\det F}\left(-\frac{1}{2}C_{\gamma\gamma}G_{\alpha\beta} + F_{\beta\alpha}\right) + K[(\det(F))^2 - 1]G_{\alpha\beta}, \tag{B3}$$

where

$$C_{\alpha\beta} \equiv \delta_{\alpha\beta} + 2E_{\alpha\beta}. \tag{B4}$$

We have chosen this model because it is stable under compression, i.e., its potential energy diverges as $\det F \to 0$ [?]. As opposed to linear models [?] and to the elastic model of ??, this model proves to yield a stable dynamics for elastic body as it is compressed by bulk fluid, see below and Fig. 4. The boundary conditions for ?? are

$$\boldsymbol{u}_\mathrm{E}(\mathbf{x}_\mathrm{r}) = 0 \text{ on } \partial\Omega^\mathrm{r}_{\odot}, \tag{B5}$$

$$S_{\mathrm{E}\,\alpha\beta}(\boldsymbol{u}^t_\mathrm{E})\big|_{\mathbf{x}_\mathrm{r}=\boldsymbol{\xi}(s)}\,\epsilon_{\beta\gamma}\frac{\mathrm{d}\xi^\gamma}{\mathrm{d}s} = \sigma^\mathrm{c}_{\mathrm{F}\,\alpha\beta}(\boldsymbol{\varphi}^t(\boldsymbol{\xi}(s)))\epsilon_{\beta\gamma}\frac{\mathrm{d}\varphi^{t,\,\gamma}(\boldsymbol{\xi}(s))}{\mathrm{d}s} \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}. \tag{B6}$$

In the left-hand side of ??, the quantity $\epsilon_{\beta\gamma}\frac{\mathrm{d}\xi^\gamma}{\mathrm{d}s}$ represents the unit normal to $\partial\Omega^\mathrm{r}_{\bigcirc}$, and similarly for the right-hand side.

Finally, the equations of motion for bulk fluid domain are ????, with boundary conditions ???? and

$$\boldsymbol{u}^t_\mathrm{D}(\mathbf{x}_\mathrm{r}) = \boldsymbol{u}^t_\mathrm{E}(\mathbf{x}_\mathrm{r}) \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}, \tag{B7}$$
$$\dot{\boldsymbol{u}}^t_\mathrm{D}(\mathbf{x}_\mathrm{r}) = \dot{\boldsymbol{u}}^t_\mathrm{E}(\mathbf{x}_\mathrm{r}) \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}. \tag{B8}$$

?? and ?? ensure that the displacement and velocity of elastic body and bulk fluid domain are conforming at the elastic body-bulk fluid domain interface, and they replace ????, respectively.

2. Dynamics

The system dynamics is obtained as follows: given ${\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}}^{n-1}$, ${\varsigma^\mathrm{r}_\mathrm{F}}^{n-3/2}$, $\boldsymbol{u}^{n-1}_\mathrm{E}$, $\dot{\boldsymbol{u}}^{n-1}_\mathrm{E}$, $\dot{\boldsymbol{u}}^{n-1}_\mathrm{D}$ and $\dot{\boldsymbol{u}}^{n-1}_\mathrm{D}$ from the preceding step, we

- **Update elastic body.** Solve for $\boldsymbol{u}^n_\mathrm{E}$ and $\dot{\boldsymbol{u}}^n_\mathrm{E}$ the discrete version of ??

$$\rho_\mathrm{E}\frac{\dot{u}^{n,\,\alpha}_\mathrm{E} - \dot{u}^{n-1,\,\alpha}_\mathrm{E}}{\Delta t} = \frac{\partial S_{\mathrm{E}\,\alpha\beta}(\boldsymbol{u}^n_\mathrm{E})}{\partial \mathsf{x}^\beta_\mathrm{r}}, \tag{B9}$$
$$\frac{\boldsymbol{u}^n_\mathrm{E} - \boldsymbol{u}^{n-1}_\mathrm{E}}{\Delta t} = \dot{\boldsymbol{u}}_\mathrm{E}, \tag{B10}$$

with boundary conditions obtained from ????:

$$\boldsymbol{u}^n_\mathrm{E} = 0 \text{ on } \partial\Omega^\mathrm{r}_{\odot}, \tag{B11}$$

$$\epsilon_{\beta\gamma}\; S_{\mathrm{E}\,\alpha\beta}(\boldsymbol{u}^n_\mathrm{E})\big|_{\mathbf{x}_\mathrm{r}=\boldsymbol{\xi}(s)}\frac{\mathrm{d}\xi^\gamma}{\mathrm{d}s} = \sigma^\mathrm{r}_{\mathrm{F}\,\alpha\beta}(\boldsymbol{\xi}(s); {\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}}^{n-1}, {\varsigma^\mathrm{r}_\mathrm{F}}^{n-3/2}, \boldsymbol{u}^n_\mathrm{E})\epsilon_{\beta\gamma}\; F_{\gamma\delta}(\boldsymbol{u}^{n-1}_\mathrm{E})\big|_{\mathbf{x}_\mathrm{r}=\boldsymbol{\xi}(s)}\frac{\mathrm{d}\xi^\delta}{\mathrm{d}s} \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}. \tag{B12}$$

In ??, we have used Eqs. (2) and (3) to rewrite the derivative of $\boldsymbol{\phi}(\boldsymbol{\xi}(s))$.

- **Update bulk fluid domain.** Solve for $\boldsymbol{u}^n_\mathrm{D}$ and $\dot{\boldsymbol{u}}^n_\mathrm{D}$ the boundary-value problems given by ???? with boundary conditions ??, ?? and

$$\boldsymbol{u}^n_\mathrm{D} = \boldsymbol{u}^n_\mathrm{E} \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}, \tag{B13}$$
$$\dot{\boldsymbol{u}}^n_\mathrm{D} = \dot{\boldsymbol{u}}^n_\mathrm{E} \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}. \tag{B14}$$

- **Update bulk fluid.** Solve the boundary-value problems given by ???? and boundary conditions ???????????? and

$$\overline{\boldsymbol{v}}_{\mathrm{F}} = \dot{\boldsymbol{u}}_{\mathrm{E}}^{n} \text{ on } \partial\Omega^{\mathrm{r}}_{\mathbf{O}}, \tag{B15}$$

which replaces ?? for a rigid body, and solve ??.

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